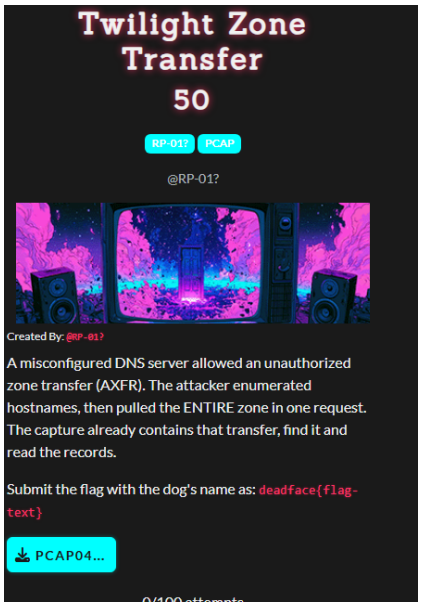
Solving “Twilight Zone Transfer” (50 pts - PCAP)

A network-forensics walkthrough of recovering a flag from an unauthorized DNS AXFR zone transfer.



Challenge page used during the solve.

1. The Challenge

The capture contained reconnaissance followed by a full DNS zone transfer. The task was to locate that transfer and read the returned DNS records.

2. Find the DNS Traffic

The UDP packets show repeated hostname enumeration against the same DNS server. The important change is a later TCP connection to port 53, because AXFR transfers commonly use TCP.

3. Identify the AXFR Request

```
Client: 10.10.14.9:50280
Server: 192.168.56.10:53
Query type: AXFR (0x00fc)
Zone: deadface-cc.net
```

4. Read the Zone Records

The server returned the complete zone in one TCP DNS response. Among the A, NS, and TXT records was a deliberately named TXT record for _flag.deadface-cc.net.

```
_flag.deadface-cc.net TXT "deadface{dns_zone_transfer}"
```

5. The Flag

```
deadface{dns_zone_transfer}
```

6. General Method

• Filter for DNS first, then look for TCP/53 when AXFR is suspected.

• Decode query type 252 (AXFR).

• Inspect all returned TXT records; CTF flags are often placed there to demonstrate the impact of a misconfigured zone transfer.

Summary

Field	Result
Category	PCAP / network forensics
Protocol	DNS over TCP
Vulnerability	Unauthorized AXFR zone transfer
Flag	deadface{dns_zone_transfer}